IAP Safety Planner P0-P5 □□□□□□ Debug □□□□

> □□□□ Safety Planner P0-P5 □□□□□□□□ Codex □□□□□□□□□□□□□□ debug □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□topic □□□□□□□□□□□□□□□□□□□□□□□□□□□□

---

0. □□□□

0.1 Codex □□□□

1. □□□□ `all_degraded_lidar_good` □□□□□□ Phase 0 -> Phase 8 □□□
2. □□□□ run □□□□□ `test_planner_manifest.json`□validator summary□rosbag□□□□□ summary□
3. □□□□□□□□□□□□□□□□□□□□□□□ `Next branch`□□□□□□□□ Phase□
4. □□□□□□□ launch □□□□ synthetic/manual □□□□□□□□□□□ `BLOCKED_SCENARIO_MISSING`□□□□□□□□□□□□
5. Pass/fail □□□□□□□□□□□□□□□□□ warning□□□□□□ hard fail□

0.2 □□□□

□□	□□
`PASS`	□□ hard pass criteria □□□validator/analyzer □□□□
`FAIL`	hard fail criteria □□□□□□□□□ debug□
`BLOCKED_SCENARIO_MISSING`	□□□□□□□□□□□□□□□□□/□□□□□
`INCONCLUSIVE`	bag/topic/CSV □□□□□□□□□□□□□□□□□□□□□

0.3 □□□□□□

```
cd /home/dev/ws_iap
source /opt/ros/jazzy/setup.bash
source install/setup.bash
```

□□□□□□ build □□□□□□

```
colcon build --packages-select iap --event-handlers console_direct+
ctest --test-dir build/iap -R "test_(risk_grid_map|predictor_module|unified_risk_grid|future_pl_field_predictor)" --
output-on-failure
ctest --test-dir build/ego_planner -R
"test_(p0_risk_grid_runtime|p5_runtime_integrity_gate|p2_candidate_ranking|p3_reference_bias|planning_risk_context)"
--output-on-failure
ctest --test-dir build/bspline_opt -R test_p1_integrity_cost --output-on-failure
ctest --test-dir build/path_searching -R test_p4_risk_astar --output-on-failure
```

0.4 Glossary

Term	**Definition**
Query-aligned fixture evidence	Evidence that the actual P5 `queryPredictedPL()` result, not only analyzer geometry, returns the injected fixture PL for future samples inside the fixture tau/spatial window.

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1. □□□□□topic □□□

1.1 Launch □□□□

□□□□□□□□□□□□□

```
ros2 launch iap test_planner.launch.py \
  experiment:=<experiment> \
  run_duration_s:=<seconds> \
  validation_duration_s:=<seconds> \
  start_rviz:=false \
  run_validator:=true \
  record_bag:=true \
  <overrides...>
```

□□□□□□ RViz □□□□

```
start_rviz:=true
```

1.2 □□ topic □□ profiles

□□□ run □□□□□□□□□□ profile □□□`timeout` □□□□ Codex □□□

Profile	□□	□□□□
`T_BASE`	`timeout 8 ros2 topic hz /iap/integrity`	□□□□□□nominal □ 10 Hz□
`T_LIDAR`	`timeout 8 ros2 topic hz /sim/drone_0/lidar_body`	□□□□□□□ 10 Hz□
`T_TRAJ`	`timeout 8 ros2 topic hz /drone_0_planning/bspline`	planner □□□□□□□□□□baseline □□□□□□□
`T_P0_HEALTH`	`timeout 5 ros2 topic echo /planning/risk_grid_health --field data`	JSON □□□□□ `ready/stale/valid_ratio/unknown_ratio/reason`□
`T_P0_RVIZ`	`timeout 8 ros2 topic hz /iap/rviz/predicted_pl_cloud && timeout 8 ros2 topic hz /iap/rviz/risk_validity_cloud`	P0 RViz cloud □□□□
`T_P5_STATUS`	`timeout 5 ros2 topic echo /planning/integrity_gate_status --field data`	JSON □□□□□ `phase/action/reason/future_min_im`□
`T_P5_RVIZ`	`timeout 8 ros2 topic hz /iap/rviz/p5_gate_status && timeout 8 ros2 topic hz /iap/rviz/current_traj_integrity_colored`	P5 RViz marker □□□□
`T_P1`	`test -s <export_dir>/planner_p1_integrity_cost_debug.csv`	CSV □□□□□□
`T_P2`	`test -s <export_dir>/planner_p2_candidate_ranking_debug.csv`	CSV □□□□□□
`T_P3`	`test -s <export_dir>/planner_p3_reference_bias_debug.csv`	CSV □□□□□□
`T_P4`	`test -s <export_dir>/planner_p4_risk_astar_debug.csv`	CSV □□□□□□

1.3 □□□□□□□□

□□□□ run □□□□□

```
src/iap/results/planner_validation/exports/test_planner_<experiment>_<scenario>_<stamp>/
  test_planner_manifest.json
  test_planner_validation_summary.json
  test_planner_integrity_validation.csv
  planner_p1_integrity_cost_debug.csv      # P1 □□
  planner_p2_candidate_ranking_debug.csv   # P2 □□
  planner_p3_reference_bias_debug.csv      # P3 □□
  planner_p4_risk_astar_debug.csv          # P4 □□

src/iap/results/planner_validation/bags/test_planner_<experiment>_<scenario>_<stamp>/
  metadata.yaml
  *.db3 / rosbag storage
```

□ `B0-4` □□□□□□□ run □ bag/CSV □□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
<export_dir>/figures/<experiment>_scenario_topdown.png
<export_dir>/figures/<experiment>_topic_activity_timeline.png
<export_dir>/figures/<experiment>_integrity_source_timeline.png
```

□□ run □ bag □□□□□□□□ topic□

```
/iap/integrity
/sim/drone_0/truth_odom
/drone_0_visual_slam/odom
/drone_0_planning/bspline
/drone_0_planning/pos_cmd
/ublox_driver/range_meas
/planning/risk_grid_health
/planning/integrity_gate_status
/iap/rviz/risk_grid_health
/iap/rviz/predicted_pl_cloud
/iap/rviz/risk_validity_cloud
/iap/rviz/trajectory_integrity_samples
/iap/rviz/current_traj_integrity_colored
/iap/rviz/p5_gate_status
/iap/rviz/p5_current_im_bars
/iap/rviz/p1_integrity_samples
```

```
/iap/rviz/p1_integrity_push_vectors
/iap/rviz/p1_integrity_metrics
/iap/rviz/p2_candidate_trajectories
/iap/rviz/p3_reference_bias
/iap/rviz/p4_astar_guides
```

1.4 □□□□□□□

□□□□□□ IAP log/analyzer □□□□□□□□□□□□□□□□□□□□□□□ launch export□rosbag □□□ debug CSV□□□□□□□ summary□CSV □□□□□□□□□□□ debug □□□

□□□□□□□□□□□□□□□

```
python3 src/iap/scripts/dev_planner/analyze_safety_planner_run.py \
  --experiment-id <ID> \
  --export-dir <export_dir> \
  --bag-dir <bag_dir> \
  --baseline-export-dir <baseline_export_dir> \
  --baseline-bag-dir <baseline_bag_dir> \
  --fail-on-threshold
```

□□□□□ `scripts/dev_planner/analyze_planner_p0_phase1.py` □□□□□□□

```
<export_dir>/metadata/safety_planner_analysis_summary.json
<export_dir>/csv/p0_risk_grid_health.csv
<export_dir>/csv/p0_pl_cost_distribution.csv
<export_dir>/csv/p0_reason_histogram.csv
<export_dir>/csv/p5_status.csv
<export_dir>/csv/p5_action_timeline.csv
<export_dir>/csv/baseline_vs_module_metrics.csv
<export_dir>/figures/p0_health_timeline.png
<export_dir>/figures/p0_pl_cost_distribution.png
<export_dir>/figures/p0_reason_histogram.png
<export_dir>/figures/p5_action_timeline.png
<export_dir>/figures/trajectory_overlay.png
```

`safety_planner_analysis_summary.json` □□□□□

```
{
  "experiment_id": "P0-1",
  "passed": true,
  "failures": [],
  "warnings": [],
  "next_debug_branch": "continue_to_P0-2",
  "topic_health": {},
  "p0_summary": {},
  "p5_summary": {},
  "module_metrics": {},
  "artifacts": {
    "csv": [],
    "figures": []
  }
}
```

1.5 □□□□□□

□□	**Hard fail**
Topic	□□ topic □□□□bag □ topic□JSON □□□□□
Validator	`test_planner_validation_summary.json` □ `passed=false`□
P0 normal	`ready=false`□`stale=true`□`valid_ratio` □□ `<=0.6`□□□□ `valid_ratio=0 && unknown_ratio=1`□
P0 fallback	`unknown_ratio` □□ `reason` □□□□□□□□ `ok`□
P0 reason	`provider_stale_count/provider_invalid_count/occupied_skip_count` □□□ reason/histogram □□□□□
P5 normal	nominal/open-sky □□□□ emergency□□□ replan storm□final gate fail□
P5 unsafe	□□ unsafe □□□□□ `REQUEST_REPLAN` □ `REQUEST_EMERGENCY_STOP_CANDIDATE`□
P1 metrics-only	`applied_to_objective=true` □□□□□□□ baseline□
P2 metrics-only	winner □□□□□

□□	**Hard fail**
P3 coverage	coverage □□□□□ global bias□
P4 no-collision	□ collision segment □□□ P4 guide□

2. □□□□□□□

□□□□□□□□□□□□ preset □□□□□□□□□□□□□□□□□□□□□□□□ `BLOCKED_SCENARIO_MISSING`□

□□	□□□□	□□□□□
Synthetic risk field provider	P0-5□P1-4□P2-2□P3-1□P3-4	□□□ affine field□risk block□two-lane/two-corridor field□
High-risk zone injection	P5-3□P5-4□P5-7	□□□□□□□□□□□□□ `PL > AL` □ `IM < emergency_margin`□
GNSS/integrity pause or delay	P5-5□R-2□R-3	□□□ `/iap/integrity` □ GNSS epoch 2s □□□
PL NaN/local unknown injection	R-4□R-5	□□□□□ NaN□invalid□low valid ratio□
P4 collision-guide fixture	P4-2□P4-3□P4-6	□□□□□ collision segment□□□ original guide □ risk-aware guide □□□
CPU/latency stress injection	R-1□R-7□R-9	□□□ P0 refresh□□□ predictor latency□□□ replan storm□

3. □□ Debug Ladder

Codex □□□□□□□□ 5 □□□□□ fail □□□□□□□□

Step	□□	□□	□□□
L0 build	□ 0.3 build/test □□	□□□□□□□□□	fail -> □ build/unit□
L1 baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_fused_nominal_off run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	baseline validator pass□□ P5 action	fail -> Phase 0 debug□
L2 P0 open sky	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0 ready/stale/unknown □□	fail -> Phase 1 debug□
L3 P5 normal	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=gnss_open_sky run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P5 □□□□ emergency/final fail	fail -> Phase 2 debug□
L4 module smoke	□□□ P1/P2/P3 preset 60s	debug CSV □□□□ hard fail	fail -> □□ Phase□

4. Phase 0: Baseline Lock

□□□□□□ safety planner □□□ original EGO □□□□□□□

ID	□□□□	**Launch □□**	□□□□	**Topic □□**	□□□□	□□□□	□□□□
B0-1	fused nominal baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_fused_nominal_off run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ preset □□□□□ P0-P5 off	`T_BASE`, `T_LIDAR`, `T_TRAJ`	manifest, validator CSV/JSON, bag	`/iap/integrity` □□□□□□□□□□□□□ safety action	validator manifest P5 □□□P5 □ replan/e
B0-2	open-sky baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_fused_nominal_off scenario:=gnss_open_sky run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	`scenario:=gnss_open_sky`	`T_BASE`, `T_TRAJ`	□ B0-1	□ PL □□□□□ P0/P5 □□□□□□	validator topic □□□ □□
B0-3	corridor baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_corridor_off run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	corridor scenario□P0-P5 off	`T_BASE`, `T_LIDAR`, `T_TRAJ`	□ B0-1	□□□ original planner □□□ corridor □□□□□□□ □□□□ safety gate	safety g topic/act □□validat □□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
B0-4	fallback baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_fused_nominal_off scenario:=fallback_only run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	fallback-only baseline□safety off	`T_BASE`, `T_TRAJ`	□ B0-1	fallback source □□□□□ P0/P5 hard action	validator safety ac

---

5. Phase 1: P0 RiskGridMap / RiskGridSnapshot

□□□□□ P0 □□□□□□□□□□□□□□ P0-only □□□□□□□□

ID	□□□□	Launch □□	□□□□	Topic □□
P0-1	open-sky □ PL field	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0 on□P1-P5 off□ `p0.debug_metrics_enable=true`	`T_BASE`, `T_LIDAR`, `T_P0_HEALT` `T_P0_RVIZ`
P0-2	degraded GNSS + LiDAR good field	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky scenario:=gnss_degraded_lidar_good run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0 on□□□□□ degraded GNSS	`T_BASE`, `T_P0_HEALT` `T_P0_RVIZ`
P0-3	corridor degeneracy field	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky scenario:=lidar_corridor_degenerate run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0 on□corridor scenario	`T_BASE`, `T_LIDAR`, `T_P0_HEALT` `T_P0_RVIZ`
P0-3-control	GNSS-assisted odom with LiDAR-only P0 risk	See command below	P0 predictor `source_mode=lidar_only`□GNSS epoch policy disabled□GNSS integrity off□LiDAR integrity on	`T_BASE`, `T_LIDAR`, `T_P0_HEALT` `T_P0_RVIZ`
P0-4	fallback/unknown □□	`ros2 launch iap test_planner.launch.py experiment:=p5_fallback_unknown planner_enable_p5_runtime:=false planner_enable_p5_final:=false run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	P0 on□P5 forced off□fallback inputs	`T_BASE`, `T_P0_HEALT` `T_P0_RVIZ`
P0-5	synthetic affine field □□	`python3 src/iap/scripts/dev_planner/analyze_safety_planner_run.py --experiment-id P0-5 --export-dir <export_dir> --synthetic-only --fail-on-threshold`	□□ synthetic risk field/analyzer □□	□ ROS topic □□ analyzer □
P0-6	occupied overlap / skip	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky scenario:=manual p0.skip_occupied_voxels:=true run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	□□ occupied overlap fixture	`T_P0_HEALT` `T_P0_RVIZ`

Recommended P0-3-control command:

```
ros2 launch iap test_planner.launch.py \
  experiment:=p0_open_sky \
  scenario:=lidar_corridor_degenerate \
  use_gnss:=true \
  enable_gnss_integrity:=false \
  enable_gnss_araim:=false \
  enable_lidar_integrity:=true \
  integrity_fusion_mode:=lidar_only \
  p0.predictor.source_mode:=lidar_only \
  p0.predictor.gnss_epoch_policy:=disabled \
  p0.predictor.use_current_integrity_prior:=true \
  run_duration_s:=60 \
  validation_duration_s:=60 \
  start_rviz:=false \
  run_validator:=true \
  record_bag:=true
```

---

6. Phase 2: P5 Runtime / Final Integrity Gate

□□□□□ P5 □□□ hard safety gate□□ action/reason □ PL/AL/IM □□□□□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□
P5-1	open-sky □ □□□□□	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=gnss_open_sky run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0+P5□□□□	`T_BASE`, `T_P0_HEALTH`, `T_P5_STATUS`, `T_P5_RVIZ`	P5 status CSV, action timeline	action `OK` □□□final fail 0□emergency 0	er st fa
P5-2	degraded □□□ debounce	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=gnss_degraded_lidar_good run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P5□ degraded input	`T_BASE`, `T_P0_HEALTH`, `T_P5_STATUS`	P5 status/action	`bad_ratio==0`□ `future_min_im` □□ □□□predicted AL □□ □□□ warning/replan□ current IM □□□□□□□ □ emergency storm	□□ ba fu fir ur va
P5-3	future high-risk zone -> replan	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=manual run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true p5.pred_alert_limit_mode:=current_msg_constant`	□□ high-risk zone injection	`T_P5_STATUS`, `T_P5_RVIZ`	P5 trajectory samples and query-aligned fixture evidence	□□ 1s □□□ `PL > AL` □□□ `REQUEST_REPLAN`	□ fu fu sa qu in
P5-4	near-risk -> emergency candidate	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=manual run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true p5.future_emergency_margin_m:=0.0`	□□ near-risk injection	`T_P5_STATUS`, `T_P5_RVIZ`	P5 status, samples	`first_bad_tau <= emergency_time_s` □ action □□ emergency candidate	□ ba
P5-5	current integrity stale	`ros2 launch iap test_planner.launch.py experiment:=p5_fallback_unknown run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ integrity delay/pause □ fallback stale	`T_P5_STATUS`	stale duration timeline	stale □□ replan□□□ □ emergency candidate	st er de
P5-6	future unknown field	`ros2 launch iap test_planner.launch.py experiment:=p5_fallback_unknown run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	P0+P5 fallback unknown	`T_P0_HEALTH`, `T_P5_STATUS`	reason histogram, P5 unknown timeline	unknown □□□ replan□□□ unknown □□□□	ur re
P5-7	final gate reject	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor scenario:=manual run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true planner_enable_p5_final:=true`	□□ rejected trajectory fixture	`T_P5_STATUS`, `T_TRAJ`	final gate fail timeline, bspline publish timeline	unsafe new trajectory □ final gate □□□□□□	re tr cc

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□
P5-8	P5 disabled no effect	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor planner_enable_p5_runtime:=false planner_enable_p5_final:=false run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P0 on□P5 forced off	`T_BASE`, `T_P0_HEALTH`, `T_TRAJ`	manifest, trajectory	P5 status □□□□□□□ □□ P5 □□	P5 pl ov

7. Phase 3 v2: P1 Integrity Soft Cost

□□□□□□□□□□□□ P0 snapshot □□□ P1 □□□□□□□ preference□ □□□□□/□□□□□ P5 hard gate□□□ Phase 3 v1 `run_p1_2_campaign.py` □□□□□□c31/c32/c38 □□ P1-2 `BLOCKED` □□□ formal analyzer □□□□ 0 □□□□□v2 □□□□ future fresh campaign□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□
P1-1	Off/metrics-only □□□	future v2 harness□□□□□□ snapshot □ P1 off/metrics-only pair	`applied_to_objective=false`	`T_BASE`, `T_P0_HEALTH`, `T_P1`	□□□□□□□□□base objective□ support/provenance	□□□□□□□□□□□ objective □□□
P1-2	□□□□□□□	□□□□□□□10 □□□□ B-spline □□□ affine□□□□□□ flat-null□□□ risk-island □□□□	reference/enabled □ initial hash□ snapshot□□□□□□support signature	`T_P1`, `T_P0_HEALTH`	before/after objective□PL/IM□ gradient/displacement	□ null raw obje □□□min IM □□□ `gradient·dis` mirror □□□□□n
P1-3	□□□□□ sweep	future v2 harness□□□□□□□ v1 campaign	raw lambda `1e-6,3e-6,1e-5,3e-5,1e-4`□normalized reference `1e-5`□budget `0.01,0.03,0.05,0.10,0.20,0.30`□ clip `0.05,0.1,0.2`	`T_P1`	scaling/safety/feasibility/gradient-ratio summary	raw scaling □□
P1-4	Fresh □□□□	future v2 harness□□ 10 □ flat-null pair□□ 10 □□□□ sweep □ counterbalanced seed	□□ fresh□□□□□□□□□□□□□ 60 s	`T_BASE`, `T_P1`, `T_TRAJ`	null tolerance□paired result□ latency	mean PL/min I □□ exact sign
P1-5	Unknown/stale □□	future v2 harness□partial□ □ unknown□ stale□NaN	stale timeout `0.5/1.0/2.0 s`□□□ low/medium/high unknown penalty	`T_P0_HEALTH`, `T_P1`	fallback□penalty□NaN/crash evidence	unknown □□□□ fallback
P1-6	P5 □□□□	query-aligned future-risk □□□ P1 off/on + P5 on	P1 preference□P5 runtime/final on	`T_P1`, `T_P5_STATUS`	P1/P5 causal timeline	unsafe traject replan/reject

8. Phase 4: P2 Candidate Ranking

□□□□□ P2 □□□□□□□□□□□□□□□□ hard safety□□ double count P1□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
P2-1	metrics-only winner □□	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good p2.metrics_only:=true run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P2 metrics only	`T_P0_HEALTH`, `T_P2`	P2 CSV	`selected` □ original □□□ score □□□	winner □□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
P2-2	□ attempt/snapshot □ fork ranking	future fresh P2 campaign□fork primary/mirror/null/soft □□□□	P2 enabled□ □□□ planning attempt□ □□□□□□□ □ P0 snapshot	`T_P2`, `T_P0_RVIZ`	candidate identity□ snapshot□ score decomposition	primary/mirror □ □□□□□□□□□null no-op□soft □□□□ □□	identity/sr □□□selecte □□ fallbacl
P2-3	P1 off + P2 on	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good p2.metrics_only:=false planner_enable_p1:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P1 off	`T_P2`	P2 CSV	candidate_score = original optimizer score + integrity score	□□ P1 tota score □□□
P2-4	P1 on + P2 on double-count check	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good p2.metrics_only:=false planner_enable_p1:=true p1.metrics_only:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P1+P2	`T_P1`, `T_P2`	P1/P2 CSV	P2 □□ `original_cost`□ □□□□ P1 □ `total_cost`	P2 score □ total_cost count
P2-5	low valid ratio fallback	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good scenario:=fallback_only p2.metrics_only:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	low valid ratio	`T_P0_HEALTH`, `T_P2`	fallback reason CSV	P2 fallback original ranking□reason □□	low valid □ reason □
P2-6	single candidate no-op	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good manager/use_distinctive_trajs:=false p2.metrics_only:=false run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	□□□□ distinctive off	`T_P2` □ CSV □ □	P2 CSV/manifest	□□□□□□□□□ single-candidate fallback	□□□□□ cra □□

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9. Phase 5: P3 Reference Bias

□□□□□ P3 □□ coverage □□□□□ local/global reference□□□□□□□ global planner□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□	□□□□
P3-1	local high-risk target bias	`ros2 launch iap test_planner.launch.py experiment:=p3_corridor planner_enable_p3_local:=true planner_enable_p3_global:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P3 local only	`T_P0_HEALTH`, `T_P3`	P3 CSV, bias marker	local target □ coverage □□□ risk □ □	target □□/□□ □□reason □	local bias □ □
P3-2	local no coverage fallback	`ros2 launch iap test_planner.launch.py experiment:=p3_corridor`	local coverage □□	`T_P0_HEALTH`, `T_P3`	fallback reason CSV	fallback nominal target	coverage □□ □□□	fallback reason □□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□	□□□□

		`scenario:=fallback_only`
		`planner_enable_p3_local:=true`
		`planner_enable_p3_global:=false`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
		`validator_require_gnss_valid:=false`
		`validator_require_lidar_valid:=false`						

P3-3	no-backtracking	`ros2 launch iap`
		`test_planner.launch.py`
		`experiment:=p3_corridor`
		`scenario:=manual`
		`planner_enable_p3_local:=true`
		`planner_enable_p3_global:=false`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
			□□ low-risk-behind fixture	T_P3	P3 CSV/RViz	□□□ risk □ □□	□□ backtracking	no-backtrackin pass

P3-4	global two-corridor bias	`ros2 launch iap`
		`test_planner.launch.py`
		`experiment:=p3_corridor`
		`scenario:=manual`
		`planner_enable_p3_local:=false`
		`planner_enable_p3_global:=true`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
			□□ two-corridor field	T_P3, T_P0_RVIZ	corridor coverage heatmap	global waypoints □□□ risk corridor	coverage □□ □□ bias□□□□ risk corridor	selected □ detour □□□

P3-5	insufficient corridor coverage	`ros2 launch iap`
		`test_planner.launch.py`
		`experiment:=p3_corridor`
		`scenario:=fallback_only`
		`planner_enable_p3_global:=true`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
		`validator_require_gnss_valid:=false`
		`validator_require_lidar_valid:=false`
			coverage □□	T_P0_HEALTH, T_P3	P3 CSV	P3-global □□□□ fallback	coverage □□ □□ global bias	fallback reason □□

P3-6	excessive detour fallback	`ros2 launch iap`
		`test_planner.launch.py`
		`experiment:=p3_corridor`
		`scenario:=manual`
		`planner_enable_p3_global:=true`
		`p3.max_detour_ratio:=1.2`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
			□□ excessive-detour fixture	T_P3	P3 CSV	detour □□ □ fallback original reference	detour □□□□ selected	fallback pass

P3-7	biased waypoint planning fail fallback	`ros2 launch iap`
		`test_planner.launch.py`
		`experiment:=p3_corridor`
		`scenario:=manual`
		`planner_enable_p3_global:=true`
		`run_duration_s:=90`
		`validation_duration_s:=90`
		`start_rviz:=false`
		`run_validator:=true record_bag:=true`
			□□ planGlobal fail fixture	T_P3, T_TRAJ	P3 CSV, FSM events	biased planning fail □ fallback original	fail □□ fallback □ crash	fallback pass

10. Phase 6: P4 Risk-aware Local A* Guide

□□□□□ P4 □□ collision segment A* guide □□ risk preference□□□ original obstacle hard rejection□timeout □ fallback□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
P4-1	no collision segment no-op	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide scenario:=gnss_open_sky run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P4 on□□ collision	`T_P0_HEALTH`, `T_P4` □ CSV □	P4 CSV	`p4_call_count=0` □ no-collision fallback	□ collision □□ path
P4-2	collision segment + two guides	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide scenario:=manual p4.lambda_p4_risk:=0.05 p4.max_extra_path_ratio:=1.3 run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ collision-guide fixture	`T_P4`, `T_P0_RVIZ`	P4 CSV/RViz	selected guide □ path ratio □□□ mean/max risk	□□ risk query□ selected □ risk guide
P4-3	risk path too long fallback	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide scenario:=manual p4.max_extra_path_ratio:=1.05 run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ long-detour fixture	`T_P4`	P4 CSV	risk path □□ fallback original guide	□ ratio □ selected
P4-4	snapshot unavailable fallback	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide p0.enable_risk_grid:=false run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	P4 requested, P0 off	`T_TRAJ`, `T_P4` □ CSV □	P4 CSV/manifest	original A* fallback□□ crash	P4 □□ null snapshot crash□ hard fail
P4-5	unknown risk fallback/penalty	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide scenario:=fallback_only run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	unknown field	`T_P0_HEALTH`, `T_P4`	P4 CSV	unknown_count □ □□□ policy fallback □ penalty	unknown □□ low risk
P4-6	occupied hard rejection	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide scenario:=manual run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ occupied-low-risk fixture	`T_P4`, `T_P0_RVIZ`	P4 CSV	occupied voxel □□ low risk □ hard reject	risk preference □□□

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11. Phase 7: Integrated Ablation

□□□□□ P0-P5 □□□□□□ baseline □□□□□□□□□ collision □ planner instability□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
A-0	off baseline	`ros2 launch iap test_planner.launch.py experiment:=baseline_corridor_off run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	all safety off	`T_BASE`, `T_TRAJ`	baseline metrics	□□□	validator fail

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
A-P0	P0 only	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky scenario:=lidar_corridor_degenerate run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0 only	`T_P0_HEALTH`	P0 metrics	□□□□□□ baseline	P0 □ trajectory health fail
A-P5	hard gate only	`ros2 launch iap test_planner.launch.py experiment:=p5_corridor run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P5	`T_P0_HEALTH`, `T_P5_STATUS`	P5 metrics	unsafe execution □□□□ □□□□□□□ risk path	hard gate fail [replan storm
A-P1	soft cost only	`ros2 launch iap test_planner.launch.py experiment:=p1_degraded_lidar_good p1.metrics_only:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P1	`T_P1`	P1 metrics	mean risk □□	risk □□□ feasibility fail
A-P2	candidate ranking only	`ros2 launch iap test_planner.launch.py experiment:=p2_degraded_lidar_good p2.metrics_only:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P2	`T_P2`	P2 metrics	low-risk candidate □□	winner □□□□
A-P3	reference bias only	`ros2 launch iap test_planner.launch.py experiment:=p3_corridor run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P3	`T_P3`	P3 metrics	target/reference □□ risk	coverage/fallba □□
A-P4	local A* guide only	`ros2 launch iap test_planner.launch.py experiment:=p4_manual_collision_guide run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P4	`T_P4`	P4 metrics	□ collision guide case □□	no-collision □ path
A-P1P5	preference + hard gate	`ros2 launch iap test_planner.launch.py experiment:=p1_degraded_lidar_good planner_enable_p5_runtime:=true planner_enable_p5_final:=true p1.metrics_only:=false run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	P0+P1+P5	`T_P1`, `T_P5_STATUS`	combined metrics	mean risk □□P5 violation □	P1/P5 □□□ final fail □□
A-P123	planner preference stack	`ros2 launch iap test_planner.launch.py experiment:=p1_degraded_lidar_good planner_enable_p2:=true planner_enable_p3_local:=true planner_enable_p3_global:=true p1.metrics_only:=false p2.metrics_only:=false run_duration_s:=90`	P0+P1+P2+P3	`T_P1`, `T_P2`, `T_P3`	stack metrics	risk □□□□□□□□□□	double count□ detour □□□ fallback □□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□	□□□□
		`validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`					
A-ALL	full system	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good run_duration_s:=120 validation_duration_s:=120 start_rviz:=false run_validator:=true record_bag:=true`	P0-P5 all	all relevant topic checks	full ablation report	min IM □□□ violation □□ collision □□	collision□ emergency storm□runtime □

12. Phase 8: Robustness / Stress

□□□□□□□□□□□□□□□□□□□□fallback/reason/action □□□□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□
R-1	P0 refresh □□	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good p0.refresh_period_s:=1.0 run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	slower P0	`T_P0_HEALTH`, `T_P5_STATUS`	runtime profile	P0 age □□□ stale □□□□
R-2	predictor/GNSS pause	`ros2 launch iap test_planner.launch.py experiment:=p5_fallback_unknown run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ GNSS pause injection	`T_P0_HEALTH`, `T_P5_STATUS`	reason timeline	`stale_gnss_epoch/no_gnss_` □□□P5 replan/escalate
R-3	`/iap/integrity` delay	`ros2 launch iap test_planner.launch.py experiment:=p5_fallback_unknown run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ integrity delay injection	`T_BASE`, `T_P5_STATUS`	stale duration CSV	current stale duration □□ acti
R-4	PL field local NaN	`ros2 launch iap test_planner.launch.py experiment:=p0_open_sky scenario:=manual run_duration_s:=60 validation_duration_s:=60 start_rviz:=false run_validator:=true record_bag:=true`	□□ NaN injection	`T_P0_HEALTH`, `T_P0_RVIZ`	invalid/unknown CSV	NaN -> unknown/invalid reas
R-5	low valid ratio	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good scenario:=fallback_only run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true validator_require_gnss_valid:=false validator_require_lidar_valid:=false`	low valid ratio	all relevant checks	fallback histograms	P1/P2/P3/P4 fallback□P5 replan/unknown policy
R-6	sudden high-risk block	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good scenario:=manual run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ dynamic risk block	`T_P0_HEALTH`, `T_P5_STATUS`	event timeline	P5 replan□□□□ preference □□
R-7	high CPU / latency	`ros2 launch iap test_planner.launch.py`	□□ CPU stress	all checks	latency profile	□□□ fallback□□□□

ID	□□□□	Launch □□	□□□□	Topic □□	□□□□	□□□□
		`experiment:=all_degraded_lidar_good run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	wrapper			
R-8	start/goal outside risk field	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good p0.size_x_m:=10.0 p0.size_y_m:=10.0 run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	shrink P0 coverage	`T_P0_HEALTH`, `T_P5_STATUS`, `T_P3`	out-of-map CSV	P3 fallback□P5 unknown/repl
R-9	high-frequency replan	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true p5.bad_tick_to_replan:=1 p5.good_tick_to_clear:=1`	aggressive debounce	`T_P5_STATUS`, `T_TRAJ`	replan timeline	replan □□□□ storm
R-10	consecutive final gate fail	`ros2 launch iap test_planner.launch.py experiment:=all_degraded_lidar_good scenario:=manual planner_enable_p5_final:=true run_duration_s:=90 validation_duration_s:=90 start_rviz:=false run_validator:=true record_bag:=true`	□□ repeated final fail fixture	`T_P5_STATUS`, `T_TRAJ`	final fail CSV	retry budget □ emergency candidate □ safe fallback

13. Debug □□□

□□□□	□□□□	□□□□	□□□
`/planning/risk_grid_health` □□□	manifest `p0.enable_risk_grid`□ `p0.debug_metrics_enable`	P0 □□□□ debug □□	□ preset/override □□□ P0-1□
P0 □□ unknown	reason histogram□provider counters	stale GNSS/integrity□provider invalid□occupied skip	□ dominant reason □□□□ □ topic hz □ snapshot age□
P0 □□□□□□□	PL/cost distribution	PL □□□□color scale □□cost mapping aggressive	□□ P0-1/P0-2/P0-3 □□□□□ □ unknown fail□
P5 nominal emergency	P5 status JSON `future_min_im/first_bad_tau/pred_al_*`	AL □□□PL field □□□trajectory samples □□	□□ P5-1 with RViz□□□ AL provider□
P1 □□□	`grad_ratio`□`applied_to_objective`	lambda □□□metrics-only □□□□ risk query miss	□ P1-3 sweep□
P2 winner □□□□	P2 CSV score decomposition	original_cost/total_cost □□□ valid ratio fallback □	□ P2-3/P2-4□
P3 □□□	coverage/detour/no-backtracking □□	coverage gate □□□candidate scoring □	□ P3-2/P3-3/P3-6□
P4 □ collision □□□	P4 CSV segment_id/call count	collision trigger □□	□ P4-1□□ trigger□

14. □□□□

□□□□□□□□□□□□□□□□□□□

1. **P0** □□□□□□□P0-1 □ P0-6 □ health□reason□PL/cost□occupied/unknown □□□□□□
2. **P5 hard gate** □□□P5-1 □ P5-8 □ action/reason/final gate □□□□□□
3. **Preference modules** □□□□□P1/P2/P3/P4 □□□□□□□□ risk metrics□□□□□ P5 hard safety□

Hard pass:

```
baseline safety off □□□□
P0 unknown/stale/out-of-range □□□ 0 risk
P5 final gate rejected trajectory □□□
P1/P2/P3/P4 □□ hard safety
P2/P1 metrics-only □□□□□
P3/P4 coverage/collision gate □□
all enabled □□□ collision□□□□ nominal emergency storm
```